

LEONARDO NIETO - ENGINEERING PHYSICS

vnietobe@student.ubc.ca | [LinkedIn](#) | [Portfolio](#) | [Github](#)

PROFILE

UBC Engineering Physics student eager to gain practical experience through hardware and software engineering internships. Combines strong software knowledge with hands-on experience in machining, soldering, and electronics. Demonstrates quick learning and problem-solving through independent projects that expand technical skills, including robotics, coding, and prototyping. Personal hobbies include working out and jazz music.

SKILLS

Mechanical & Design Tools: Machining, SolidWorks, Technical Drawings, Waterjet Cutting
Programming & Software: C++, C, Python, Java, MATLAB, HTML/CSS, Git, Microsoft Office
Electronics & Instrumentation: Soldering, Arduino, Circuit Analysis, Oscilloscopes

TECHNICAL EXPERIENCE

Engineering Technician Co-op Intern

March 2023 – April 2023

SMPC Technologies - Burnaby, BC

- **Soldered printed circuit boards and thermocouple wires**, reliably achieving high-precision connections used in temperature sensing for power systems.
- **Performed quality control** on electronic components using multimeters and measurement tools, ensuring compliance with design tolerances and preventing defective parts from entering production.
- **Designed and assembled** a 20 kW power test unit, applying machining, and problem-solving skills to build a functional product.
- **Calibrated and updated** electronic power units using D-Tool software, processing 100+ units to meet client requirements.
- **Reclaimed reusable components** from decommissioned units, contributing to company recycling and cost-saving efforts.
- Strengthened **communication, responsibility, and problem-solving** through daily collaboration in a professional engineering environment.

TECHNICAL PROJECTS

Coder – VEX U Robotics Competition

July 2025 – Present

SWARM Robotics Team - Vancouver, BC

- Developed and implemented **odometry algorithms** to improve robot positioning and movement accuracy.
- Refactored legacy code into **organized classes and methods using C++ OOP**, enhancing maintainability and reducing errors.
- **Optimized software** to support competition performance, enabling efficient communication with mechanical and electrical subteams.
- Strengthened **collaboration, problem-solving, and software engineering skills** in a team-based, competitive environment.

Machining Project – Phone Holder

August 2025

UBC Student Machining Shop Course

- Designed and manufactured a custom **aluminum phone holder**, applying machining strategies to produce a precise finished product.
- Applied **lathe** and **milling** techniques, along with **waterjet cutting and tool grinding**, to fabricate and finish components to specification.
- Interpreted **technical drawings** and used **measurement tools** to ensure dimensional accuracy and product quality.
- Strengthened **independent problem-solving and machining skills** through hands-on project work.
- Strengthened communication, responsibility, and problem-solving through daily collaboration in a professional engineering environment.

Hangman Game – Python

July 2025 – Present

Programming Project

- Independently developed a **feature-rich Hangman game** using Python and Pygame, themed around the Old West, incorporating a dynamic scoring system.
- Implemented **object-oriented programming principles** to create organized, maintainable code and a polished user experience.
- Integrated **custom graphics, animations, and sound effects** to deliver a fully interactive gameplay experience.
- Published the complete project and code on [GitHub](#), demonstrating project completion and code sharing best practices.

EDUCATION

Engineering Physics, BSc. (in-progress)

UBC, Vancouver

Key Courses: Linear Circuits; Principles of Software Construction; Experimental Techniques

ADDITIONAL EXPERIENCE

Founder & President – High School's Jazz Club

Sept 2023 – June 2024

Heritage Woods Secondary School - Port Moody, BC

- Founded and grew a **student jazz club to 50+ members**, establishing a platform for weekly practice, learning, and performance.
- Organized and led **weekly rehearsals and school performances**, coordinating logistics and musical programming for club members.
- Managed **recruitment, promotion, and event planning**, rapidly expanding membership and visibility within the school community.
- Led preparation for the **2024 Surrey Schools International Jazz Festival**, resulting in a successful regional performance.
- Recognized with **Top Jazz Musician of the Year** and the **2024 Leadership Award**, demonstrating impact in leadership, teamwork, and music performance.